Mean

1 Describe what the mean measures for a set of scores.

2 Find the mean of the following data sets:

a $8, 15, 6, 27, 3$ **b** $22.4, 25.4, 19.1, 24.3, 7.4$

3 The mean of four scores is 21. If three of the scores are $17, 3$ and 8, find the fourth score.

4 Determine whether each of the following data sets has a mean of 3:

a $8, 4, 2, 3, 1$ **b** $3, 2, 5, 1, 4$ **c** $1, 3, 7, 5, 2$ **d** $2, 4, 5, 4, 3$

5 Consider the following of each statements:

a Write the notation for the sample mean.

b Write the notation for the population mean.

6 Find the sum of a set of 10 scores with a mean of 4.

7 The mean of a set of scores is 35 and the sum of the scores is 560. Calculate the number of scores in the set.

8 A set of five numbers has a mean of 10. Two of the numbers are 6 and 13. Determine whether the following 3 other numbers could be in the set:

a $15, 11, 8$ **b** $10, 13, 8$ **c** $13, 5, 6$ **d** $10, 6, 18$

9 Five numbers have a mean of 7. If 4 of the numbers are $10, 10, 8$ and 7 and the last number is x, find the value of x.

10 Han wants to try out as a batsman for a cricket team. In his last three matches, he scored $61, 75$ and 66 runs. In his last match before trying out, he wants to lift his mean to 70. If x is the number of runs he needs to score to achieve this, find the value of x.

11 The five numbers $16, 16, 17, 24, 17$ have a mean of 18. If a new number is added that is bigger than 24, will the mean be higher or lower?

12 The five numbers $11, 13, 9, 13, 9$ have a mean of 11. If a new number is added that is smaller than 9, will the mean be higher or lower?

13 The mean of a set of 41 scores is 18.6. If a score of 71.8 is added to the set, find the new

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mean. Round your answer to two decimal places.

14 A teacher calculated the mean of 25 students' marks to be 64. A student who later completed the assessment got a mark of 55. Find the new mean of the class, correct to two decimal places.

15 Over a 12-game season, Jack scored the following points:

$$12,\ 12,\ 18,\ 9,\ 12,\ 9,\ 11,\ 15,\ 15,\ 19,\ 15,\ 18$$

a Find the population mean.

b In three games against his arch rivals, he averaged 10. Is μ bigger or smaller than $\overline{x}$?

16 If $\mu = 30$, determine whether the following could be represented by μ:

a The mean weight, in kilograms, of all the people attending a concert.

b The mean age of 100 people, chosen randomly from the crowd, at a concert.

c The mean age of all the people attending a concert.

d The mean weight, in kilograms, of 100 people, chosen randomly from the crowd, at a concert.

17 The table shows the scores of Student A and Student B in five separate tests:

Test	Student A	Student B
1	97	78
2	87	96
3	94	92
4	73	72
5	79	86

a Find the mean score for Student A.

b Find the mean score for Student B.

c What is the combined mean of the scores of the two students.

d What is the highest score overall? Which student obtained that score?

e What is the lowest score overall? Which student obtained that score?

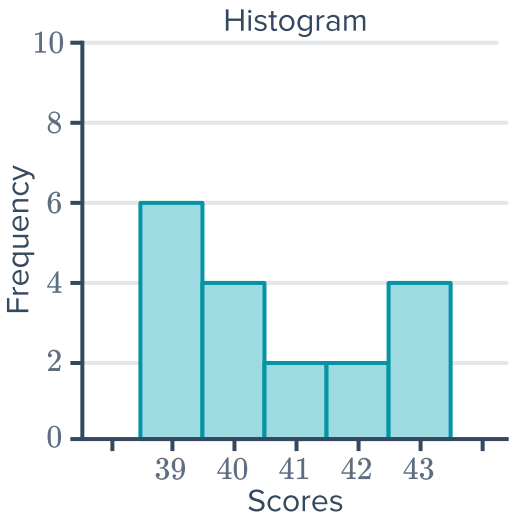
18 Consider the following histogram:

a Find the total number of scores.

b Calculate the sum of the scores.

c Calculate the mean, correct to two decimal places.

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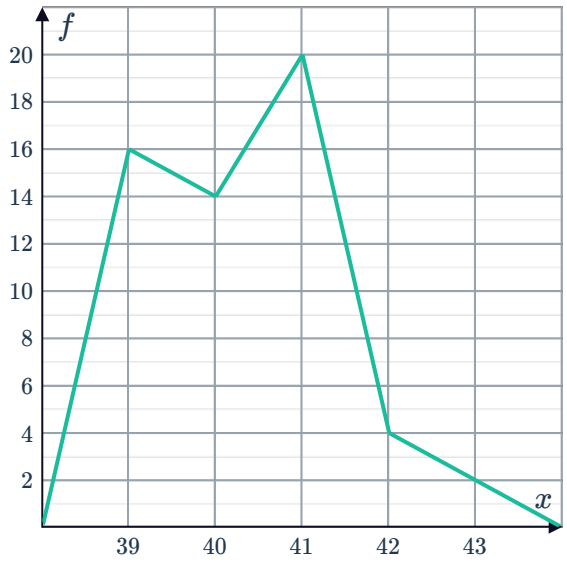


19 From the frequency polygon shown:

a Find the number of scores.

b Calculate the sum of the scores.

c Calculate the mean of the scores, correct to two decimal places.



20 The following stem plot shows the batting scores of two cricket teams, England and India:

a What is the highest score from England?

b What is the highest score from India?

c Find the mean score of England.

d Find the mean score of India.

e Calculate the combined mean of the two teams.

England		India
10	3	1 2 4 7
6 6 5 5 5 5	4	0 2 9
7 3	5	2 5
	6	4

Key: $0|2|4 = 20$ and 24

Median

21 Find the median of the following set of scores:

a 6, 8, 9, 11, 16, 17, 18

b 11, 11, 13, 14, 18, 22, 23, 25

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22 For each of following set of scores:

i Sort the scores in ascending order. **ii** Calculate the median.

a $23, 25, 13, 9, 11, 21, 24, 17, 20$

b $24, 11, 1, 3, 24, 28, 16, 16$

c $65.2, 64.3, 71.6, 63.2, 45.2, 62.2, 46.8, 58.7$

d $44.9, 45.6, -54.8, 74.7, -77.6, -42.6, 67.9, 40.6$

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23 State the position of the median in an ordered set of:

a 69 scores

b 152 scores

24 Find the median for the data in the given stem plot:

	Leaf
1	1 3
2	0 3 4
3	3 4 6 7
4	0 0 1 5
5	0 1
6	5
7	4 7
8	1 4

Key: $1|2 = 12$

25 How many scores are there in a distribution if the number of scores is even and the median lies between the 24th and 25th scores?

26 For each frequency distribution table, find the median of the scores:

a

Score	Frequency
11	24
12	9
13	21
14	9
15	6

b

Score	Frequency
23	2
24	26
25	27
26	24
27	25

c

Score	Frequency
10	6
14	9
18	6
21	7
24	8

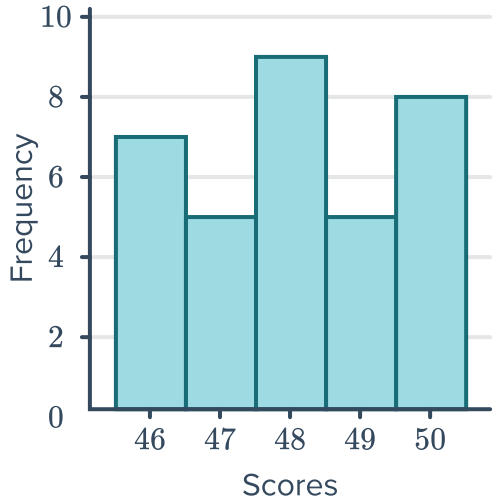
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27 For each of the following histograms:

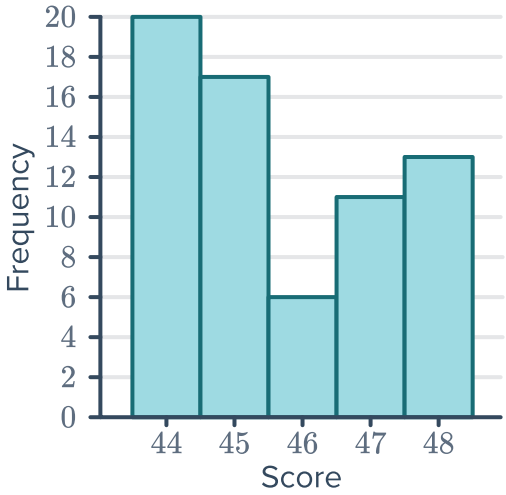
i Find the number of scores.

ii Find the median.

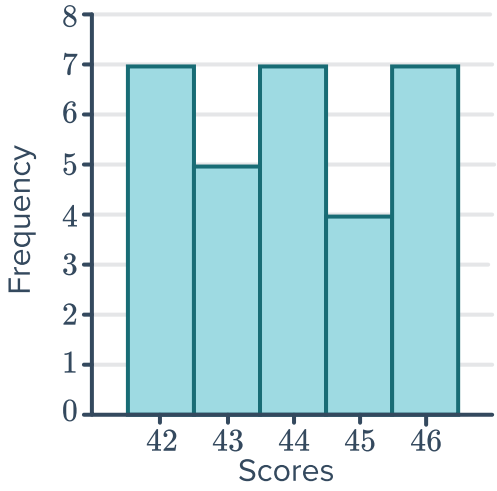
a



b



c



28 Find four consecutive odd numbers whose median is 40.

29 On Sunday, 20 planes were delayed at the airport. The dot plot shows the number of hours each departure was delayed:

a What was the median number of minutes a plane was delayed?

b What percentage of planes were delayed for longer than the median time?

c If a plane is delayed for more than 30 minutes, the airline must pay $\$5000$. In total, how much were airlines fined that day?



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30 A real estate agent wanted to determine a typical house price in a certain area. He gathered the selling price of some houses (in dollars):

$$317\,000,\ 320\,000,\ 347\,000,\ 360\,000,\ 378\,000,\ 395\,000,\ 438\,000,\ 461\,000,\ 479\,000,\ 499\,000$$

a Find the mean house price.

b What percentage of the house prices exceeded the mean?

c Find the median house price.

d What percentage of house prices exceeded the median?

31 The luggage check-in weight (in kilograms) of passengers is given in the following frequency table:

a Complete the table.

b Find the median check-in weight.

c Find the mean check-in weight, correct to two decimal places.

d When one of the passengers sees that the weight of their luggage is 21 kg, they decide to add some items to the bag. This changes the mean or the median?

Weight (x)	Frequency (f)	fx
16	12	
17	17	
18	21	
19	10	
20	18	
21	22	
Total		

Mode

32 Find the mode of the following scores:

a 8, 18, 5, 2, 2, 10, 8, 5, 14, 14, 8, 8, 10, 18, 14, 5

b 2, 2, 6, 8, 8, 8, 8, 12, 14, 14, 14, 14, 18, 18

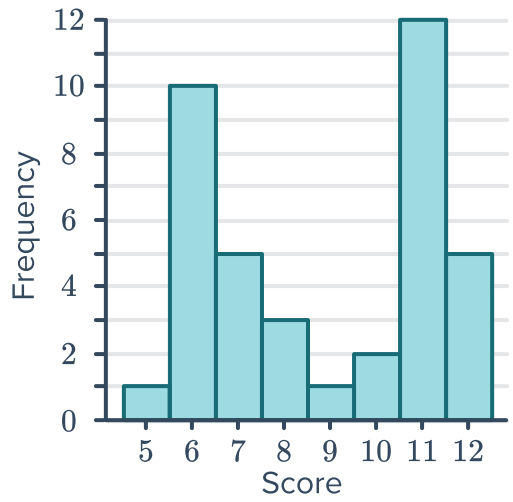
c

Score	Frequency
25	17
26	42
27	35
28	32
29	12
30	20

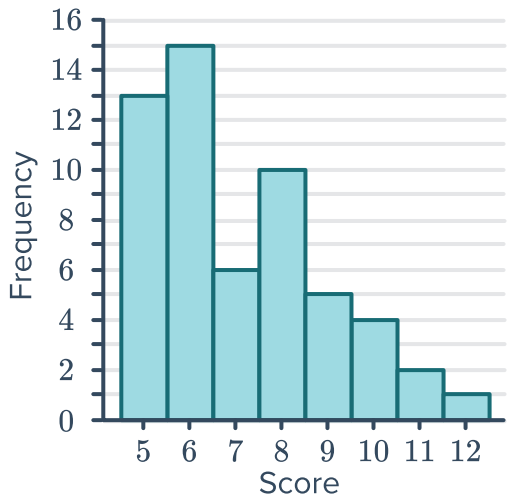
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33 Which of these bar graphs shows the higher mode?

A



B



34 Which of the following dot plots has the lowest mode?

A



B



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